Medly: A Retrieval-Augmented Generation and Recommendation Chatbot for Medical Question Answering

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***Abstract*—This work introduces an Arabic biomedical question-answering system that designed to assist non-expert users in obtaining informative medical information through a chatbot interface. The system integrates retrieval techniques and large language models (LLMs) to provide responses for users questions. The system was developed to serve Arabic users through support for Arabic questions and Arabic answers through translation modules. A large corpus of biomedical data is used to build the foundation of the system's knowledge base. A MedCPT module was adopted to encode both the biomedical corpus and user translated queries into a shared semantic vector space. FAISS, which is a fast vector similarity search library was utilized to retrieve the most relevant documents based on cosine similarity. A carefully structured prompt was engineered using these top-ranked documents, which is was then passed to the Google's large language model (Gemini) to return an informative medical response, further more the response included lifestyle health recommindations, and suggested topics for further reading. The final result is displayed to the user via our (Medly) chatbot interface that was developed using the Streamlit library. The final model was evaluated and tested using Retrieval-Augmented Generation Assessment Suite, achieving a faithfulness score of 83%.**

***Index Terms*—RAG, MedCPT, chatbot, FAISS, Cosine Similarity, Natural Language Processing (NLP), Large Language Model (LLM)**

I. Introduction

Recent progress in natural language processing (NLP) has advanced the creation of systems capable of understanding and responding to complex, domain-specific queries with high accuracy [1]. In the medical field, access to relevant and reliable biomedical information is important for patient care and medical decision-making. The challenge lies in effectively retrieving relevant biomedical information from large and complex datasets and subsequently generating clear explanations that are suitable for patients [2].

To address these challenges, retrieval-augmented generation (RAG) frameworks have grown as an effective approach. These systems first search large biomedical databases to find the most relevant documents related to the user's question. Then, they use these documents as context to generate accurate and clear answers[3]. The system feeds the relevant documents retrieved along with the user's question into a large language model, which processes this combined information to produce a response [4].

This work presents a Retrieval-Augmented Generation (RAG) based chatbot designed to answer medical questions by combining document retrieval and language generation.

A core challenge in such systems is to obtain high-quality vector representations (embeddings) of both queries and documents, which allows efficient and accurate retrieval of relevant information.

To achieve this, we use the MedCPT model, which is trained for biomedical retrieval tasks. MedCPT converts both user queries and medical documents into vector embeddings in the same semantic space. By embedding queries and documents separately, the system can quickly find the most relevant documents related to a given question using vector similarity methods [5].

Once the relevant documents are retrieved, they are passed along with the user query to a large language model. This model synthesizes the information from the documents and generates coherent and informative answers with some recommendation questions to the user.

II. Data Set

In this paper we want to utilize a large corpus of medical questions that are both relevant, widespread, have both specialized but also general health and medical advice. For that we used a dataset collected from different trusted resources, the breakdown of the corpus is shown in Table I. The corpus contains 216,102 total samples, with each entry as a question with one or more answers [6]. The largest number of samples are from MEDMCQA, this is a large-scale, Multiple-Choice Question Answering (MCQA) dataset designed to address real-world medical entrance exam questions. It contains question from different medical topics including dental, paediatrics, skin, surgery, pharmacology, radiology and more. This makes it suitable to answer very precise and correct information regarding these medical info. [7]

Source	Percentage	Number of Samples
MedMCQA	89%	192,331
MedQuAD	8%	17,288
BioASQ	2%	4,322
Kaggle	1%	2,161
Total	**100%**	**216,102**

TABLE I
BREAKDOWN OF DATA SOURCES USED IN THE CORPUS.

In our initial design we actually intended to use a full Arabic Corpus, that consisted of multiple questions and answers for medical questions. In our search however, the Arabic datasets presented challenges on their own, one of the challenges in the extensive preprocessing needed to extract the core medical info and also the different dialects of the different Arabic datasets. Unfortunately, we were also not able to find a large dataset as our current corpus. Due to these limitations and also limitations related to resources available to do extensive pre-processing and embedding of the Arabic dataset, we opted to use this dataset in English, with Arabic support through end-to end translation.

III. WORKFLOW

This system was built mainly using the RAG (Retrieval-Augmented Generation) framework, which we specifically designed for biomedical question answering. The system architecture and complete workflow is shown in 1. It begins by transforming the data set into embeddings , which are dense vector representations of the textual documents. To do that MedCPT document encoder was used [27]. These embeddings are saved and loaded into a FAISS index (Facebook AI Similarity Search) which is an open source library designed for: Storing vector embeddings and performing fast nearest-neighbour search to find the most similar vectors.

When a user inputs a medical question through our Medly chatbot interface,the query is initially in Arabic. The system translates the Arabic query into English using a translation module. The query text is tokenized into chunks. These tokenized query passed through the MedCPT Query-Encoder model which is the same transformer-based encoder used for the document to ensure that both queries and documents share the same embedding space. The encoder generates a vector that represents the user's question.

After obtaining the vector representation of the user's query, the system perform a similarity search using cosine similarity over the embeddings vectors to obtain the top N relevant answers to the user's query. This retrieval process uses FAISS which is efficient for vector similarity search. After using the validation set and testing with different configurations, the model was set to retrieve the top 20 documents.

After retrieving the top-N most relevant candidate documents using FAISS, the system performs a re-ranking step to improve the accuracy of the results and obtain top K relevant answers from the first top N. This is achieved using the MedCPT Cross-Encoder — a transformer-based model fine-tuned specifically for biomedical question-answering and information retrieval. In this step, each of the N retrieved documents is paired with the user query and passed as input to the Cross-Encoder. The Cross-Encoder outputs a relevance score for each query document pair. These scores are then used to re-rank the N candidates retrieved by FAISS, and the top K results are selected.

After re-ranking the top-k retrieved documents using the MedCPT Cross-Encoder, the system divides the results into two segments: one used to construct the answer prompt, and the other for generating recommendation. For this model, it was found that using the top 4 documents for the response prompt was the best option.

For the answer generation, a structured prompt is created with the top relevant documents then passed to the Gemini language model. The prompt was engineered to be an instructional Prompt, Zero-shot, with context injection [28]. This prompt includes specific instructions for the model to act as a medical assistant, to provide answer based on the documents provided, take in mind the history of a chat, be direct and informative. The prompt that was used is shown in Figure 2.

```
def build_prompt(context, english_question):
    return f"""You are a knowledgeable, empathetic
    ↪ medical assistant.
    Use the retrieved context to answer naturally
    ↪ and conversationally,
    integrating facts smoothly. Do not mention that
    ↪ you are using documents
    or retrieved sources. Do not say "the documents
    ↪ say".
    Use ONLY the following medical context :
    ↪ {context}
    to answer the following question:
    ↪ {english_question}.

    In your answer:
    - Explain like a caretaker and be specific but
    ↪ professional
    - Make your answer informative, not short
    - Explain this for someone who does not have
    ↪ any medical background
    - Try to conclude an answer from the documents
    - Be compassionate but do not offer solutions
    """
```

Fig. 2. Python function to build a medical assistant prompt.

The remaining documents from the retrieval step, which we set to another 4, were used to construct the recommendation prompt, which will be explained later. After passing this

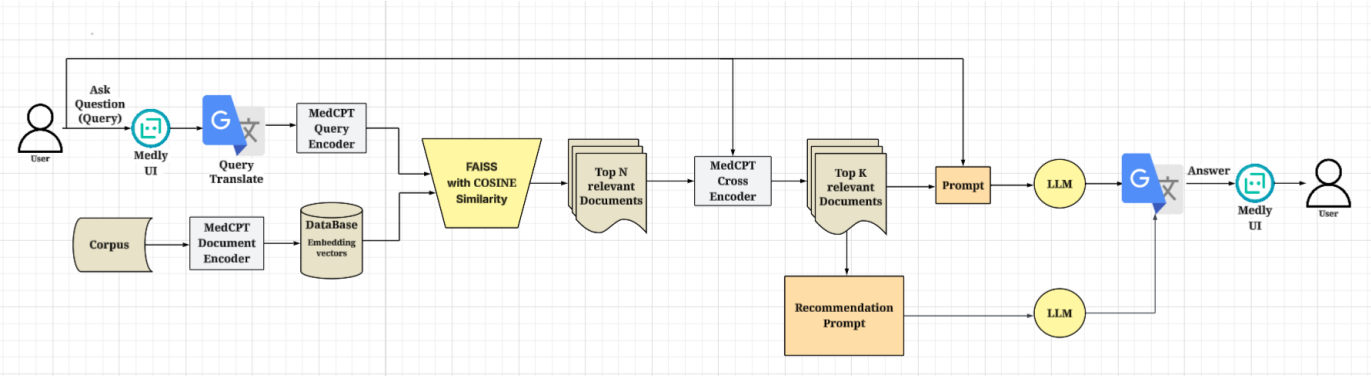


Fig. 1. System architecture of the proposed Conversational Arabic RAG model.

prompt to the LLM, a response is provided, then being translated and displayed to the user.

IV. MedCPT

In our RAG-based chatbot system, we adopt MedCPT model as the core component for embedding both user queries and medical documents into vector representations, and enabling efficient semantic search and retrieval in the medical domain. MedCPT (Medical Contrastive Pre-trained Transformer) is a retrieval model built specifically for zero-shot biomedical information retrieval. It's the first biomedical IR model that includes a pair of retriever and re-ranker closely integrated by contrastive learning. It's trained by 255M query-article pairs from PubMed search logs [13].

MedCPT includes two stages, first-stage retriever and a second-stage re-ranker. The retriever includes a query encoder (QEnc) and a document encoder (DEnc). The re-ranker is a cross-encoder that is accurate due to the cross-attention computation between query and article tokens [14].

Our system utilizes both stages of MedCPT, the first stage which has the Query Encoder (QEnc) and the Document Encoder (DEnc). The Query Encoder (QEnc) is used to convert the medical question of the user into a fixed-size embedding vector. The question is tokenized using the PubMedBERT tokenizer [15]. The tokenized input is passed through the MedCPT Query Encoder, which is a transformer model based on PubMedBERT [16]. The encoder generates a sequence of hidden states for the input tokens. The final hidden state corresponding to the [CLS] token (the first token) is extracted and used as a vector representation of the entire input. The Document Encoder (DEnc) works similarly but is used to encode biomedical documents (titles and abstracts). It takes the document as input. Then, the text is tokenized with the same tokenizer. The tokens passed through the MedCPT Document Encoder. The [CLS] token embedding is extracted to represent the full document. The second stage which is a cross-encoder, cross-encoder processes both the query vector and the document vector in one Transformer model to produce a score that represents how relevant the document is to the query. Since it is computationally expensive, we used it to re-rank the result of FAISS search.

V. Recommendation Generation

Adding a recommendation algorithm is essential to enhance the user experience instead of providing direct answers to queries. As a result, we extend the retrieval augmented generation (RAG) by adding a second stage of response generation process that generates context-based suggestions. LLMs and prompt engineering are used to accomplish this. LLMs have the potential to understand the user's query context and provide useful recommendations. [17][18]

Prompt engineering is the key strategy for enhancing goal-driven output and LLM replying, which involves developing a set of natural language instructions that the LLM model can follow to provide a response. In our work, the recommendation phase occurs after the initial factual answer has already been generated. The prompt design includes the original user query and a certain number of retrieved documents. Thy system use the top n relevant documents for generating the initial answer. Then in the recommendation phase, we use additional k documents (next best ranks) that are not used in generating the initial answer. These are specifically chosen to extend the scope available to LLM to increase the likelihood that the model would encounter information related to lifestyle, behaviour, or prevention.[19]

The prompt is designed to provide two types of output:the first is recommending healthy habits for a better lifestyle. This type is recommended Only when the user's query relates to a condition that may be prevented or treated. The second kind involves recommending related topics for the user to research. This kind is always recommended. The prompt contains strict instructions not to provide clinical or medication-related advice in order to protect user safety.

VI. Results

For Medly user interface, we utilized Streamlit, a Python-based framework that allows for development of interactive

web applications. This design a user-friendly interface that supports Arabic language input and output.

The following screenshots shown in Figure 3 and Figure 4 display Medly interface.



Fig. 3. Home page of Medly chatbot



Fig. 4. Chat page of Medly chatbot

Once the final answer is generated by the Gemini model, it is displayed to the user through the Medly chatbot interface. The response is presented in a clear and friendly format, suitable for non-expert users. The response includes a medical content. Additionally, recommendations based on the retrieved content shown below the main answer.

We tested the system using two types of medical questions: General medical questions, such as wondering about specific conditions or symptoms. Advice-seeking questions, where the user asks about lifestyle advice about a particular health topic.
The chatbot was able to provide well-structured and informative answers. In advice-seeking cases, the system give lifestyle suggested. In addition, the response included a list of related topics for further reading.

The following figures: Figure 5, Figure 6, Figure 7, show samples responses that generated by the system:

VII. Model Evaluation

One of the most challenging aspects of creating a RAG system is its evaluation. In a traditional manner, metrics like Kappa has been used, however this requires a large number of professionals to decide how relevant or irrelevant a response is.



Fig. 5. Sample one response generated by the chatbot



Fig. 6. Sample two response generated by the chatbot



Fig. 7. Sample three response generated by the chatbot

While this approach has been used in traditional IR systems, upscaling is a necessity when it comes to RAG applications. To evaluate our model, we have utilized RAGAS which stands for (Retrieval-Augmented Generation Assessment Suite). It was developed by contributors from the LangChain ecosystem. It is an open-source library designed to evaluate RAG pipelines automatically, without the need for large-scale human annotation. [23]

RAGAS also provide measures to test RAG pipeline without the need of ground truth in many cases, which can be hard to obtain. In our model, we focused on two measure: faithfulness and answer relevancy.

The Faithfulness metric measures how factually consistent a `response` is with the `retrieved context`. It ranges from 0 to 1, with higher scores indicating better consistency [24]. Mathematically it can be expressed as follows, [22] :

$$\text{Faithfulness} = \frac{\text{Number of Correct Facts in the Response}}{\text{Total Number of Facts in the Response}}$$

On the other hand, the answer relevancy metric, measures how relevant a response is to the user input. Higher scores indicate better alignment with the user input, while lower scores are given if the response is incomplete or includes redundant information [25]. For this metric the idea of reverse engineering is used, in this metric a number of questions are being generated, then being embedded , then compared with the generated embedding of the query and then being averaged. Mathematically it can be expressed as follows:

$$\text{Answer Relevancy} = \frac{1}{N}\sum_{i=1}^{N} \text{cosine similarity}(E_{g_i}, E_o)$$

$$\text{Answer Relevancy} = \frac{1}{N}\sum_{i=1}^{N} \frac{E_{g_i} \cdot E_o}{\|E_{g_i}\|\|E_o\|}$$

Where:

- E_{g_i}: Embedding of the ith generated question.

- E_o: Embedding of the user input.

- N: Number of generated questions (default is 3).

To properly test and evaluate our RAG model, we created two datasets, one for the test and one for validation. Each we compiled of 100 questions of the most asked medical and health questions from different resources in the past 5 years [26].

The validation set was helpful in building and training the model, we were able to use it to edit and test the prompt, and also as we tested different strategies, encoders and LLMs.

In our final teste set, we were able to obtain a faithfulness score of average **0.83**, which is considered high for this RAG system. While answer relevancy was not as good, it was almost **0.52**, unfortunately, the answer relevancy kept fluctuating. For many questions it was very high , for other half it was zero. There might be many reasons for this, such as the ambiguity of the query or the prompt.

But according to this metric, the fundamental idea behind answer relevancy is to demonstrate if the original question can be rebuilt entirely from the response [25]. Although this might be significant in some situations, our system was designed to put helpfulness, coherence, and domain relevance ahead of perfect query mirroring. As a result, even while the answer relevance measure is instructive, it could not match the intended user experience of our system.

VIII. Future Work

In the future work, we aim to add user profiles - such as age, gender, medical history and preferences - to enhance recommendations and personalize lifestyle suggestions based on user's conditions or preferences. Additionally, we aim to introduce a new type of recommendations focusing on recommending doctors, medical specialists and hospitals based on the the user geographical location.

A promising enhancement can be made is integrating user-feedback mechanisms for the iterative improvement of the quality and relevance of generated recommendations. This enhances personalization and allows the system to adapt to real-world usage.

As we go on developing our model, we also aim to explain further metrics and enhancements with regards to the way the model responds to the question, in the relevant context.

IX. Conclusion

In this presented work, our model Medly represented a strong example of Retrieval-Augmented Generation centred around medical questions, further more combing the idea of recommendation to create a system that cares about the well-beings. This model has successfully combined and integrated the use of information retrieval methodologies, LLMs and recommendation systems. We were able to provide Arabic answers and queries. This study presents an Arabic biomedical question-answering system aimed at helping non-expert users acquire useful medical information via a chatbot interface. The system combines retrieval methods and large language models (LLMs) to deliver answers to user inquiries. The system was designed to assist Arabic users by providing support for Arabic queries and Arabic responses via translation modules. A substantial collection of biomedical information is utilized to establish the system's knowledge base. A MedCPT module was utilized to encode the biomedical corpus and user-translated queries into a common semantic vector space. FAISS, a rapid library for vector similarity search, was employed to find the most pertinent documents by using cosine similarity. A meticulously crafted prompt was developed utilizing these highest-ranked documents, which was subsequently submitted to Google's large language model (Gemini) to generate an informative medical reply. Additionally, the response contained lifestyle health suggestions and recommended topics for further exploration. The end result is presented to the user through our (Medly) chatbot interface created with the Streamlit library. The concluding model was assessed and tested with the Retrieval-Augmented Generation Assessment Suite, obtaining a faithfulness score of 83%.

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